

Lesson 19: Nonparametric tests

Pagona TB, Chapter 13

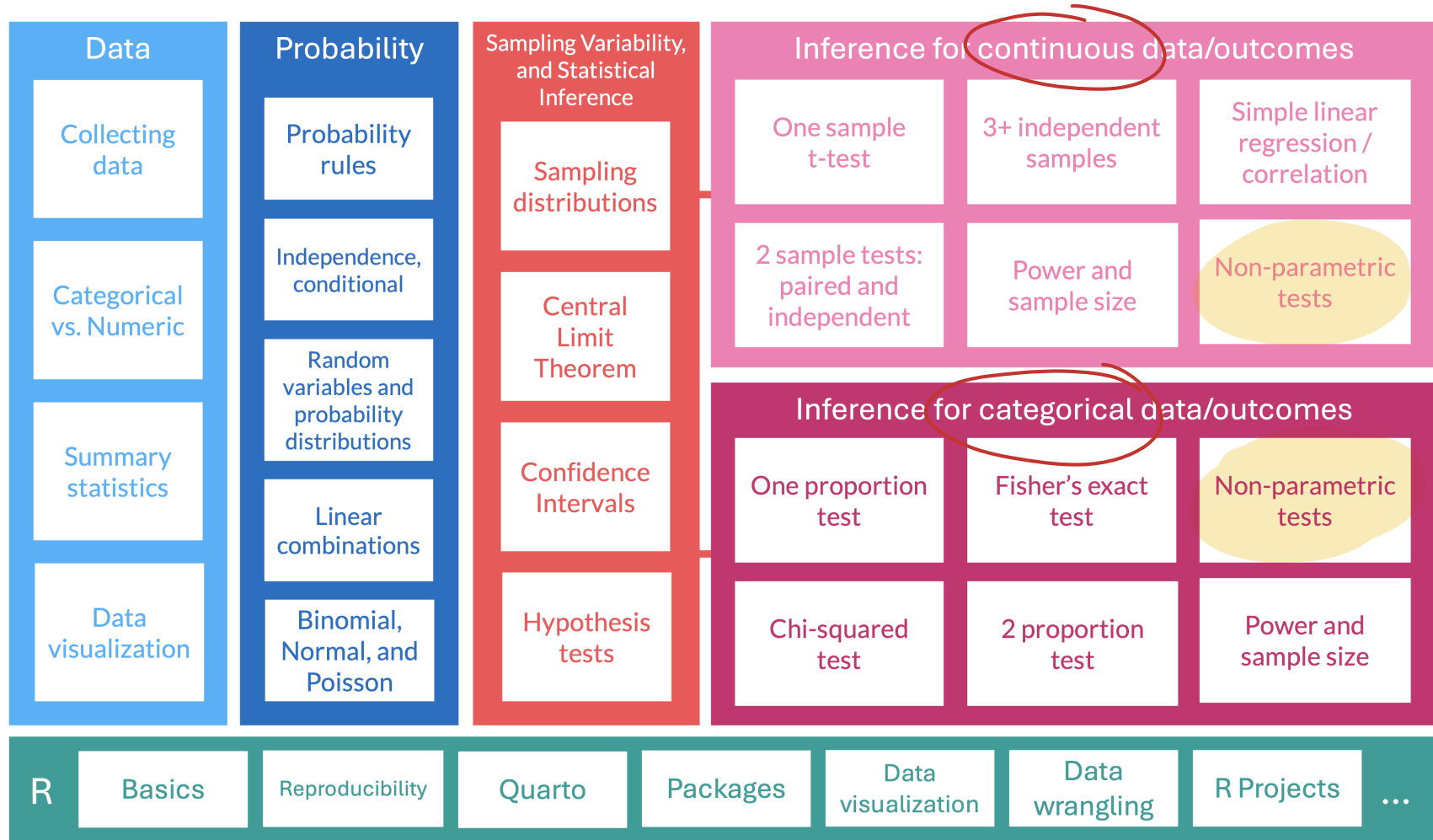
Meike Niederhausen and Nicky Wakim

2025-12-03

Learning Objectives

1. Understand the difference between and appropriate use of parametric and nonparametric tests
2. Use the (Wilcoxon) Signed-rank test to determine if a single sample or paired sample are symmetric around some value.
3. Use the Wilcoxon Rank-Sum test to compare two independent numeric samples.
4. Use the Fisher's Exact test to determine if two categorical variables are associated.
5. Use the Kruskal-Wallis test to compare two or more independent numeric samples.

Where are we?



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Background: parametric vs nonparametric

- Parametric vs. nonparametric
 - Basically saying: assuming a distribution for our data vs. not assuming a distribution for our data
- In all of our inference so far, we have assumed the population (that the data come from) has a specific distribution
 - means & prop. χ^2 test ANOVA
 - Normal distribution, T-distribution, Chi-squared distribution, F-distribution
- Each of those distribution can be **parameterized** from certain population parameters
 - For example: Normal distribution is completely described (parameterized) by two parameters: μ and σ
- Our inference and analysis was all based in the assumed distribution
 - But remember: we have specific assumptions that we need to check in order to use those distributions!
- Nonparametric procedures
 - ↳ large enough sample or $np \geq 10$
 - Make fewer assumptions about the structure of the underlying population from which the samples were collected
 - Work well when distributional assumptions are in doubt.

★ if sample small, non parametric ★

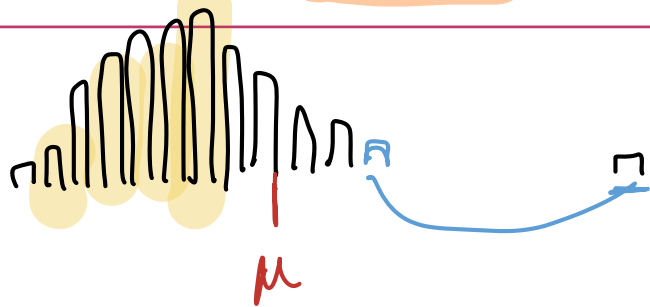
Nonparametric tests: Pros vs. cons

Pros

- Fewer assumptions
 - Can handle smaller sample sizes
 - No assumptions about the distribution of the data's population
- Tests are based on ranks
 - Therefore outliers have no greater influence than non-outliers.
 - Since tests are based on ranks we can apply these procedures to ordinal data

Cons

- Less powerful than parametric tests (due to loss of information when data are converted to ranks)
- While the test is laid out for us, it may require substantial (computer) work to develop a confidence interval
- Ties in ranks make the test harder to implement
- Some nonparametric methods can be computationally intensive, especially for large datasets or complex designs



Parametric and nonparametric tests

assumptions
not met
for
parametric

Type of data	Parametric test	Nonparametric test
Single sample, numeric	Single mean hypothesis test or t-test (L11)	Sign test or (Wilcoxon) signed-rank test
Paired sample, numeric	Mean difference (paired) hypothesis test or t-test (L12)	Sign test or (Wilcoxon) signed-rank test
Two independent sample, numeric	Difference in means hypothesis test or two sample t-test (L13)	Wilcoxon rank-sum test or Mann-Whitney U test
Single sample, binary	Single proportion hypothesis test (L15)	Binomial
<u>Two independent sample</u> , binary	Difference in proportions hypothesis test (L15)	↓
2+ independent samples, binary	Chi-squared test (L16)	Fisher's Exact test
2+ independent samples, numeric	ANOVA test or F-test (L17)	Kruskal-Wallis test

We still follow the general hypothesis test process

1. Check the **assumptions**

- We will not meet the parametric assumptions!
- There are some assumptions for the nonparametric tests



meet assumptions
⇒ parametric
do not meet
⇒ nonparametric

2. Set the **level of significance** α

3. Specify the **null** (H_0) and **alternative** (H_A) **hypotheses**

- In symbols
- In words
- Alternative: one- or two-sided?

4. Calculate the **test statistic** and **p-value**

- We will not discuss the test statistic's equation

5. Write a **conclusion** to the hypothesis test

- Do we reject or fail to reject H_0 ?
- Write a conclusion in the context of the problem

Poll Everywhere Question 1

13:20 Wed Dec 3

...

90%

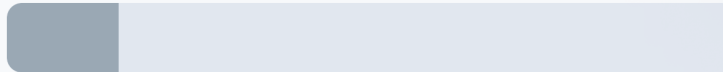


Join by Web PollEv.com/nickywakim275



Which of the following is a primary assumption of parametric tests?

The data are ordinal.



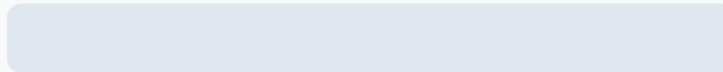
12%

The data come from a specific distribution. ✓



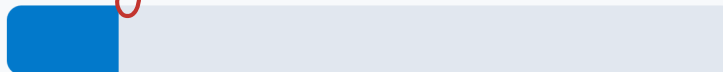
76%

The data contain no outliers.



0% ✓

The sample size must be greater than 30. ✓

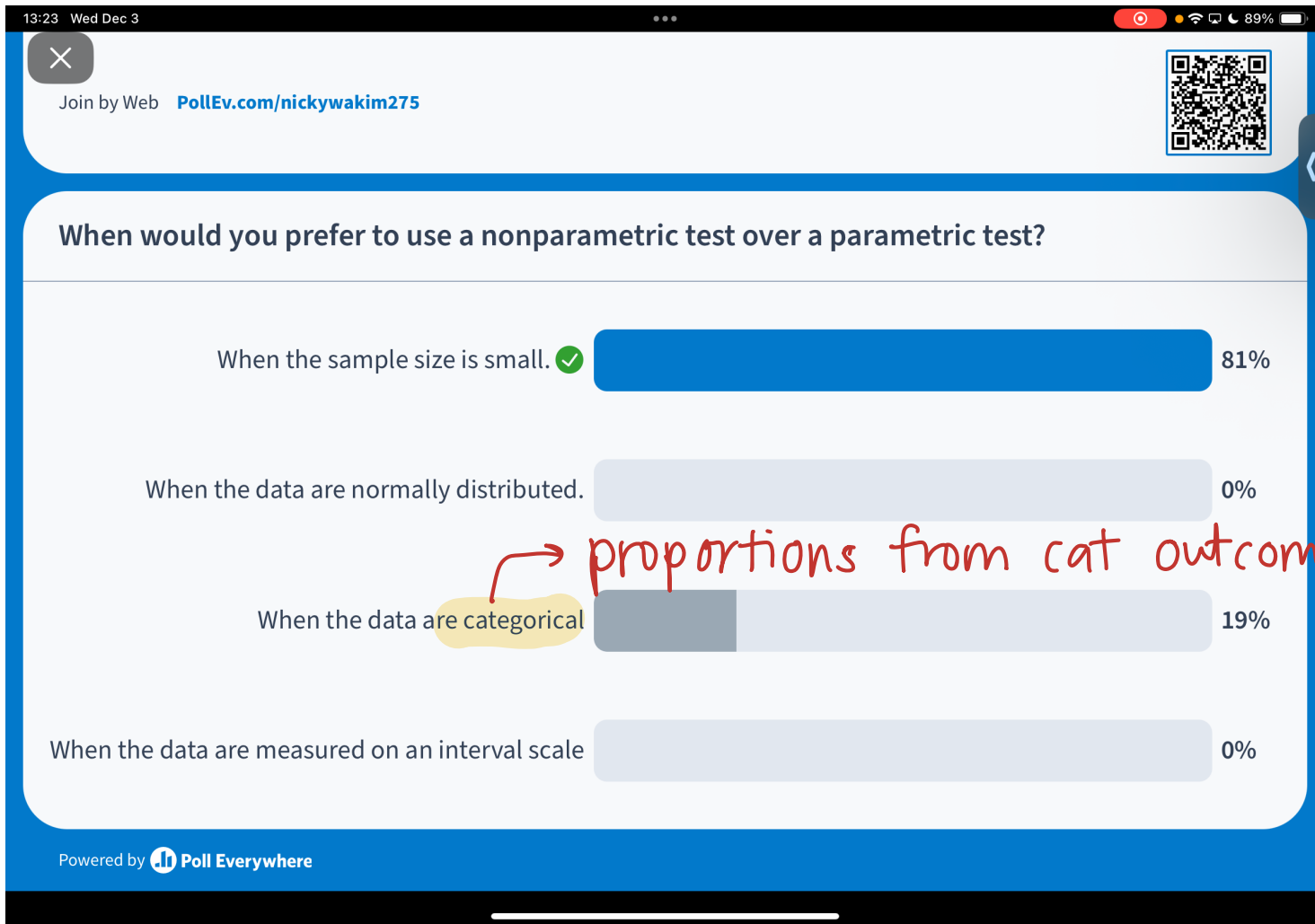


12%

is large enough

$np \geq 10$

Poll Everywhere Question 2



proportions from cat outcomes → can approx Normal (which is parametric)

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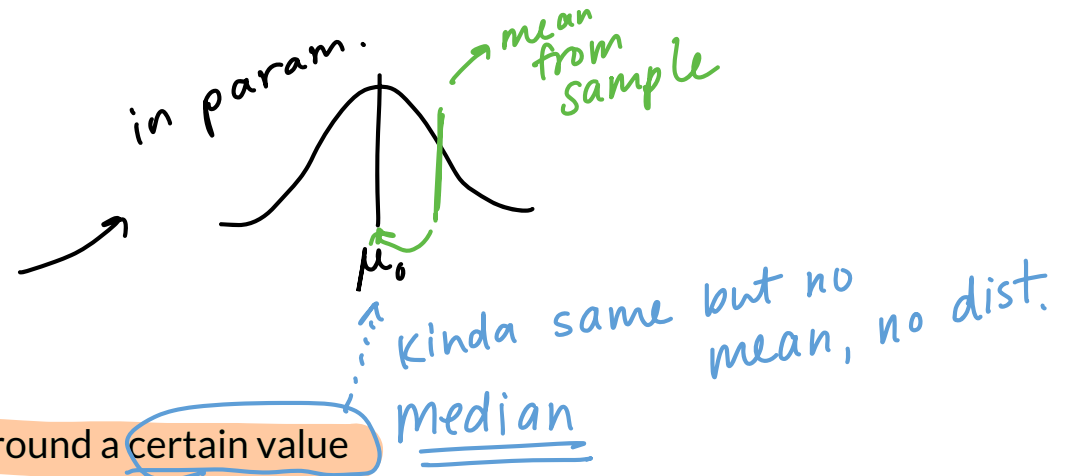
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(Wilcoxon) Signed-rank test

- The (Wilcoxon) signed-rank test is used for
 - Paired samples (i.e., a single set of differences)
 - One-sample comparison against a specified value



- If we want to see if data are symmetric (centered) around a certain value
 - For paired data, we may want to see if the data are symmetric around 0 to determine a difference
 - For one sample, we may have an idea of a median value that our data may follow
- Think back to the parametric parallel of these!
 - If we apply the body temperature example to this: We would check if the data were symmetric around 98.6
- Data do NOT need to be approximately normal

Example: Intraocular pressure of glaucoma patients

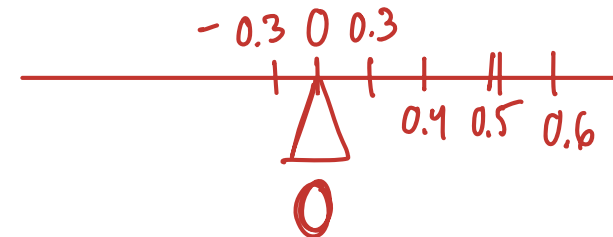
- Intraocular pressure of glaucoma patients is often reduced by treatment with adrenaline.
- A new synthetic drug is being considered, but it is more expensive than the current adrenaline alternative.
- 7 glaucoma patients were treated with both drugs:
 - one eye with adrenaline and
 - the other with the synthetic drug
- Reduction in pressure was recorded in each eye after following treatment (larger numbers indicate greater reduction)

$n \geq 30$
for t.test

$7 < 30 \Rightarrow$ need to do nonparametric

Patient	Adren	Synth	d	Sign
1	3.5	3.2	-0.3	-
2	2.6	3.1	0.5	+
3	3.0	3.3	0.3	+
4	1.9	2.4	0.5	+
5	2.9	2.9	0.0	NA
6	2.4	2.8	0.4	+
7	2.0	2.6	0.6	+

- d is the difference in reduction of pressure: Synth - Adren
- Sign is + if the difference is positive and - if the difference is negative



(Wilcoxon) Signed-rank test: Hypotheses

General wording for hypotheses

H_0 : population is symmetric around some value $\tilde{\mu}_0$

H_a : population is not symmetric around some value $\tilde{\mu}_0$

NOT the mean
(usually median)

- Even if the population has a mean/median equal to $\tilde{\mu}_0$, the test may reject the null if the population isn't symmetric, thus only use if the data (differences) are symmetric.
- If the population is symmetric
 - then the mean and median coincide,
 - thus often the null hypothesis is phrased in terms of the median (or median difference) being 0

Hypotheses test for example

H_0 : the population difference in reduction of intraocular pressure in treatment with adrenaline vs. new synthetic drug is symmetric around ~~mean~~ 0

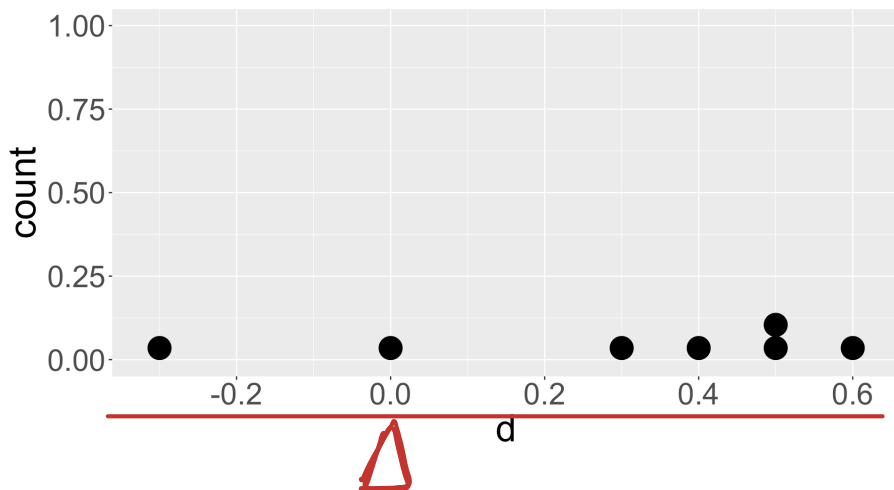
H_a : the population difference in reduction of intraocular pressure in treatment with adrenaline vs. new synthetic drug is not symmetric around ~~mean~~ 0

50% of data on one side,
50% on other

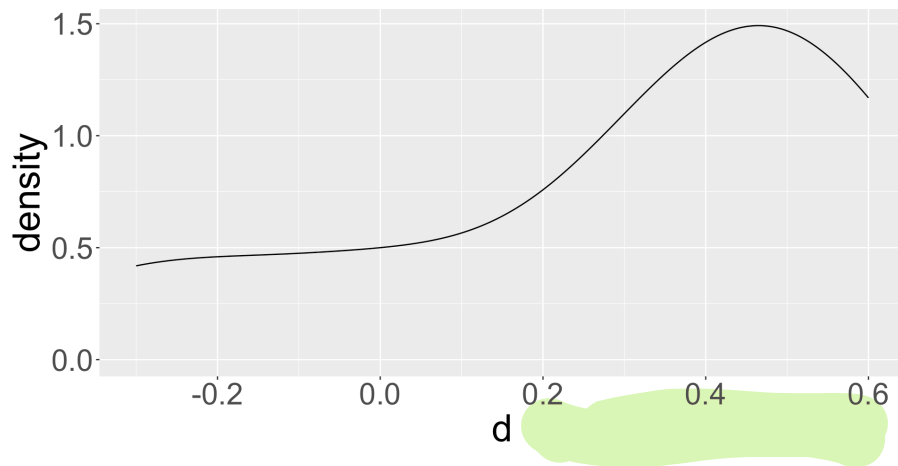
Example: Visualize the differences

Visualize the differences in reduction of pressure d : Synth - Adren

```
1 ggplot(IOP_table, aes(x = d)) +  
2   geom_dotplot() +  
3   theme(text = element_text(size = 30))
```



```
1 ggplot(IOP_table, aes(x = d)) +  
2   geom_density() +  
3   theme(text = element_text(size = 30))
```



Example: Calculate signed ranks

- Rank the **absolute values** of the differences from smallest to largest
- For ties, take the average of the highest and lowest tied ranks
 - I.e. if ranks 3-7 are tied, then assign $(3+7)/2 = 5$ as the rank
- Then calculate the **signed ranks** as +/- the rank depending on whether the sign is +/-

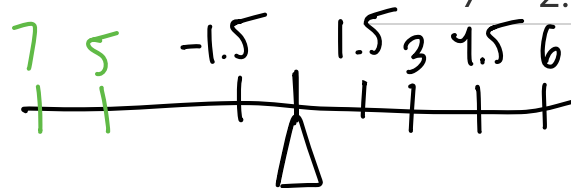
$$(1+3)/2 = 2$$

$$(1+2)/2 = 1.5$$

```

1 IOP_ranks <- IOP %>%
2   mutate(
3     Rank = c(1.5, 4.5, 1.5,
4              4.5, NA, 3, 6),
5     Signed_rank = case_when(
6       d < 0 ~ -Rank,
7       d > 0 ~ Rank
8     )
9   )
    
```

Patient	Adren	Synth	d	Sign	Rank	Signed_rank
1	3.5	3.2	-0.3	-	1.5	-1.5
2	2.6	3.1	0.5	+	4.5	4.5
3	3.0	3.3	0.3	+	1.5	1.5
4	1.9	2.4	0.5	+	4.5	4.5
5	2.9	2.9	0.0	NA	NA	NA
6	2.4	2.8	0.4	+	3.0	3.0
7	2.0	2.6	0.6	+	6.0	6.0



(Wilcoxon) Signed-rank test: Test statistic

- If the null is true:
 - The population is symmetric around some point ($\tilde{\mu}_0 = 0$, typically), and
 - The overall size of the positive ranks should be about the same as the overall size of negative ranks.
- We can split the positive and negative ranks
 - T^+ = sum of the positive ranks
 - T^- = sum of the negative ranks
- Thus, any of the following can be used as a test statistic and will lead to the same conclusion:
 - T^+ (what R is using)
 - T^-
 - $T^+ - T^-$
 - $\min(T^+, |T^-|) = T_0$

Example: calculate sums of signed ranks

- Sum of the positive ranks

$$T^+ = 1.5 + 3 + 4.5 + 4.5 + 6 = 19.5$$

- Sum of the negative ranks

$$T^- = -1.5$$

- $\min(T^+, |T^-|) = T_0 = 1.5$

Patient	Adren	Synth	d	Sign	Rank	Signed_rank
1	3.5	3.2	-0.3	-	1.5	-1.5
2	2.6	3.1	0.5	+	4.5	4.5
3	3.0	3.3	0.3	+	1.5	1.5
4	1.9	2.4	0.5	+	4.5	4.5
5	2.9	2.9	0.0	NA	NA	NA
6	2.4	2.8	0.4	+	3.0	3.0
7	2.0	2.6	0.6	+	6.0	6.0

$$T^+ - T^- = 19.5 - (-1.5) = 21$$

(Wilcoxon) Signed-rank test: Exact p-value (fyi)

- **Exact p-value** is preferable
 - This is the default method in R's `wilcox.test()`
 - if the samples contain less than 50 finite values
 - and **there are no ties**
 - *R will automatically use normal approximation method if there are ties*
 - *We will not be calculating the exact p-value “by hand.” We will be using R for this.*
-

$$p - value = 2 * P(\min(T^+, |T^-|) \leq t)$$

- t is the smaller of the calculated sums of the positive and negative ranks
- To calculate the exact p-value, we need the probability of each possible sum of ranks

(Wilcoxon) Signed-rank test in R: Glaucoma example

"Attempt" with exact p-value & running one sample test with differences

```
1 # Exact p-value
2 wilcox.test(x = IOP$d,
3             alternative = c("two.sided"),
4             exact = TRUE, correct = TRUE)
```

median of 0 (diff)

$\mu = 0$

Wilcoxon signed rank test with continuity correction

data: IOP\$d

V = 19.5, p-value = 0.07314

alternative hypothesis: true location is not equal to 0

$> 0.05 \Rightarrow$ fail to reject null

$\uparrow$ p-value $\downarrow$ evidence of alternative
 $\downarrow$ p-value $\uparrow$ evidence of alt.

(Wilcoxon) Signed-rank test: Conclusion

Recall the hypotheses to the (Wilcoxon) Signed-rank test:

H_0 : the population difference in reduction of intraocular pressure in treatment with adrenaline vs. new synthetic drug is **symmetric around $\tilde{\mu}_0 = 0$**

H_a : the population difference in reduction of intraocular pressure in treatment with adrenaline vs. new synthetic drug is **not symmetric around $\tilde{\mu}_0 = 0$**

- Significance level: $\alpha = 0.05$
- p-value = 0.07314
- Do we reject or fail to reject H_0 ? insufficient evidence of alternative

Conclusion:

There is insufficient evidence the differences in reduction in intraocular pressure ~~differs~~ between the synthetic drug and adrenaline are symmetric about 0 (2-sided Wilcoxon signed rank test p -value = 0.07314)

✓
NOT

(Wilcoxon) Signed-rank test with one sample

- One can use the (Wilcoxon) Signed-rank test when testing just one sample
- Note that we did this when in R: Ran the (Wilcoxon) Signed-rank test using just the differences
- For one sample, we can specify the null population median value:

H_0 : The population median is m

H_a : The population median is NOT m

Not-so-real example: Run (Wilcoxon) Signed-rank test for paired data with null $m = 0.7$

```
1 wilcox.test(x = IOP$d, mu = 0.7, alternative = "two.sided")
```

Wilcoxon signed rank test with continuity correction

data: IOP\$d

$V = 0$, p-value = 0.02225

alternative hypothesis: true location is not equal to 0.7

Learning Objectives

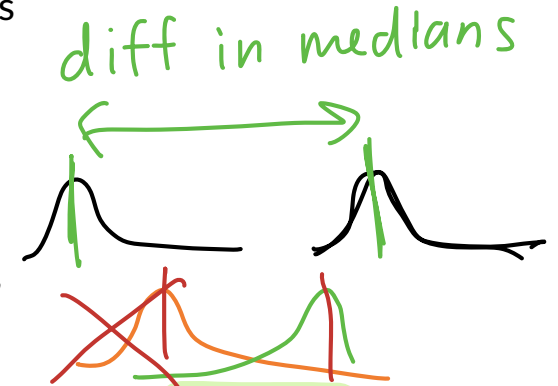
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Wilcoxon rank-sum test

- The nonparametric alternative to the two-sample t -test
 - used to analyze two samples selected from separate (independent) populations
- Also called the Mann-Whitney U test
- Unlike the signed-rank test, there is no need to assume symmetry
- Necessary **condition** is that the two populations being compared
 - have the same shape (both right skewed, both left skewed, or both symmetric),
 - so any noted difference is due to a shift in the median
- Since they have the same shape, when summarizing the test, we can describe the results in terms of a difference in medians.



Hypotheses:

H_0 : the two populations have the same median

H_a : the two populations do NOT have the same median

Example

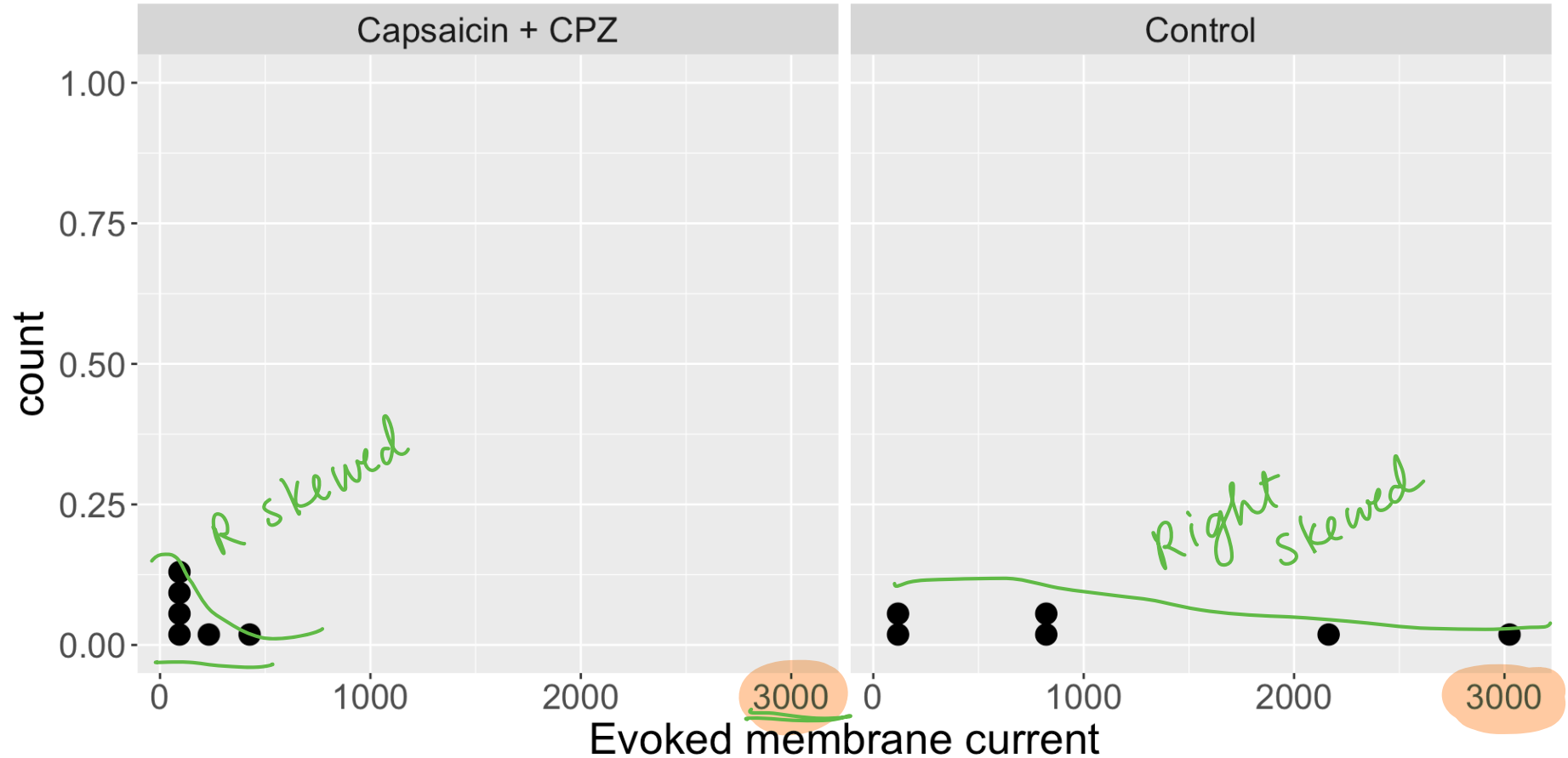
Dr. Priya Chaudhary (OHSU) examined the evoked membrane current of dental sensory neurons (in rats) under control conditions and a mixture of capsaicin plus capsazepine (CPZ). J. Dental Research 80:1518–23, 2001.

Group	variable	n	median
Capsaicin + CPZ	Memb_current	6	112
Control	Memb_current	6	822

Rat_ID	Group	Current
1	Control	3024
2	Control	2164
3	Control	864
4	Control	780
5	Control	125
6	Control	110
7	Capsaicin + CPZ	426
8	Capsaicin + CPZ	232
9	Capsaicin + CPZ	130
10	Capsaicin + CPZ	94
11	Capsaicin + CPZ	75
12	Capsaicin + CPZ	55

Example: Visualize the data

Do the independent samples have the same distribution? ✓



Wilcoxon rank-sum test: Calculating test statistic W

1. Combine the two samples together (keep track of which observations came from each sample).
2. Rank the full set of $N = n_1 + n_2$ observations.
 - If ties exist, assign average ranks to tied values (like signed-rank test)
3. Sum the ranks corresponding to those observations from the smaller sample.
 - This is a time-saving step; you could also have used the larger sample.
 - Call this sum W
4. If n_1, n_2 are both less than 10, then use an exact test (can only be done if no ties are present)
 - Otherwise use the large-sample normal approximation. *for prob Z-score*

In our example, both groups have equal n ; choose either for computing W .

$$W_{CPZ} = 1 + 2 + 3 + 6 + 7 + 8 = 27$$

$$W_{control} = 4 + 5 + 9 + 10 + 11 + 12 = 51$$

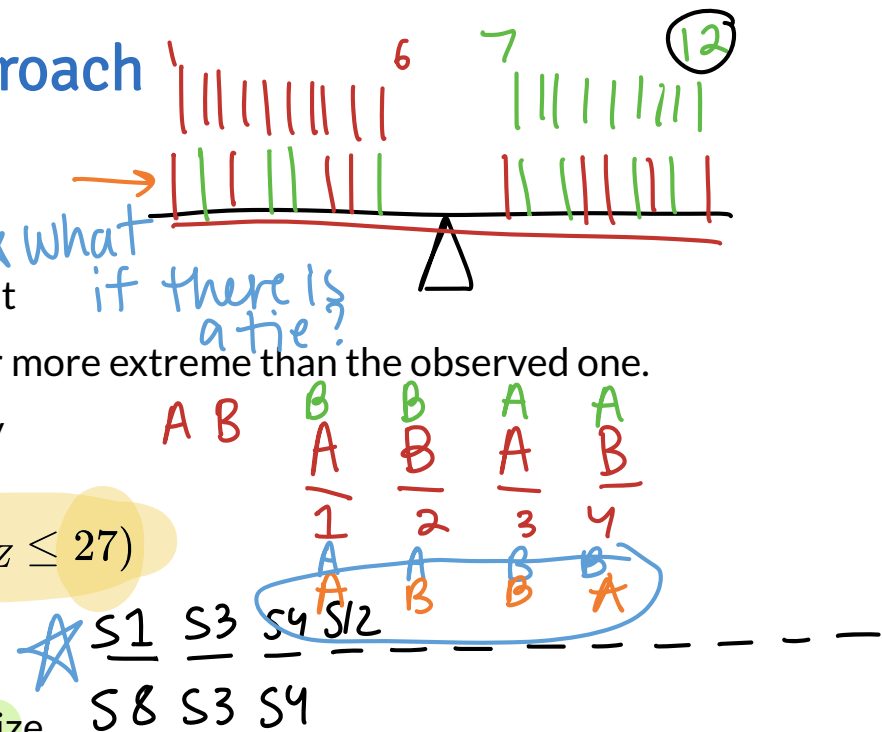
Rat_ID	Group	Current	Rank
12	Capsaicin + CPZ	55	1
11	Capsaicin + CPZ	75	2
10	Capsaicin + CPZ	94	3
6	Control	125	4.5
5	Control	125	4.5
9	Capsaicin + CPZ	130	6
8	Capsaicin + CPZ	232	7
7	Capsaicin + CPZ	426	8
4	Control	780	9
3	Control	864	10
2	Control	2164	11
1	Control	3024	12

Wilcoxon rank-sum test: Exact p-value approach

- If n_1, n_2 are both less than 10, then use an exact test,
 - otherwise use the large-sample normal approximation.
 - However, exact method can only be done if **no ties** are present
- p-value is the probability of getting a rank sum W as extreme or more extreme than the observed one.
 - Multiply the 1-tail probability by 2 for the 2-tailed probability

$$p - value = 2 \cdot P(W_{CPZ} \leq 27)$$

- To calculate $P(W_{CPZ} \leq 27)$,
 - we need to enumerate all possible sets ranks for the sample size,
 - calculate the sum of ranks for each set of possible ranks,
 - and then the probability for each sum (assuming each set of ranks is equally likely).



Wilcoxon rank-sum test: using R

Exact p-value

```
1 wilcox.test(Current ~ Group,  
2             data = CPZdata2,  
3             alternative = c("two.sided"), mu = 0,  
4             exact = TRUE)
```

Handwritten notes: "outcome" with an arrow pointing to the tilde (~) in the formula; "sample/pop" with an arrow pointing to the "Group" variable.

Wilcoxon rank sum exact test

data: Current by Group

W = 6, p-value = 0.06494

alternative hypothesis: true location shift is not equal to 0

Wilcoxon rank-sum test: Conclusion

Recall the hypotheses to the Wilcoxon rank-sum test:

H_0 : the control and treated populations have the same median

H_a : the control and treated populations do NOT have the same median

- Significance level: $\alpha = 0.05$
- p-value = 0.06494
- Do we reject or fail to reject H_0 ?

Conclusion:

There is suggestive but inconclusive evidence that the evoked membrane current of dental sensory neurons (in rats) differs between the control group and the group exposed to a mixture of capsaicin plus capsazepine (2-sided Wilcoxon rank-sum test p -value = 0.06494).

insufficient evidence that two groups
differ

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Fisher's Exact Test

- Only necessary when expected counts in one or more cells is less than 5
- Given row and column totals fixed, computes exact probability that we observe our data or more extreme data
- Consider a general 2 x 2 table:

Group	Outcome		Total
	Died Yes	Alive No	
Treatment	a ₁₂	b ₁₀₈	a+b ₁₀
Control	c	d	c+d
Total	a+c	b+d	n

- The exact probability of observing a table with cells (a, b, c, d) can be computed based on the hypergeometric distribution

$$3! = 3 \cdot 2 \cdot 1 \quad 5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$P(a, b, c, d) = \frac{(a+b)! \cdot (c+d)! \cdot (a+c)! \cdot (b+d)!}{n! \cdot a! \cdot b! \cdot c! \cdot d!}$$

- Numerator is fixed and denominator changes

Some notes on the Fisher's Exact Test

- This is always a two-sided test
- There is no test statistic nor CI
- There is no continuity correction since the hypergeometric distribution is discrete

Recall our example from Lesson 4 and 18

Question: Is there an association between age group and hypertension?

- Let's pretend that we actually had the following numbers

Table: Contingency table showing hypertension status and age group.

Age Group	Hypertension	No Hypertension	Total
18-39 yrs	2 1	10 11	12
40-59 yrs	4 4	9 9	13
60+ yrs	3 4	3 2	6
Total	9 9	22	31

Fisher's Exact test: Hypertension

1. Check expected cell counts threshold

```
1 hyp_data2 %>% expected()
```

	Hypertension	No_Hypertension
[1,]	3.483871	8.516129
[2,]	3.774194	9.225806
[3,]	1.741935	4.258065

~~We're going to pretend~~ they are less than 5.

2. $\alpha = 0.05$ ✓

3. Hypothesis test:

- H_0 : There is no association between age group and hypertension
- H_1 : There is an association between age group and hypertension

4. Calculate the p-value for Fisher's Exact test in R

```
1 fisher.test(x = hyp_data2)
```

Fisher's Exact Test for Count Data

data: hyp_data2

p-value = 0.04062

alternative hypothesis: two.sided

$0.0406 < \alpha = 0.05$

5. Conclusion to the hypothesis test

We reject the null hypothesis that age group and hypertension are not associated ($p = 0.04062$). There is sufficient evidence that age group and hypertension are associated.

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Type of data	Parametric test	Nonparametric test
Single sample, numeric	Single mean hypothesis test or t-test (L11)	Sign test or (Wilcoxon) signed-rank test
Paired sample, numeric	Mean difference (paired) hypothesis test or t-test (L12)	Sign test or (Wilcoxon) signed-rank test
Two independent sample, numeric	Difference in means hypothesis test or two sample t-test (L13)	Wilcoxon rank-sum test or Mann-Whitney U test
Single sample, binary	Single proportion hypothesis test (L15)	
Two independent sample, binary	Difference in proportions hypothesis test (L15)	
2+ independent samples, binary	Chi-squared test (L16)	Fisher's Exact test
2+ independent samples, numeric	ANOVA test or F-test (L17)	Kruskal-Wallis test

Kruskal-Wallis test: nonparametric ANOVA test

- Recall that an ANOVA tests means from 2 or more groups (often save for 3 or more)
- Conditions for ANOVA include
 - Sample sizes in each ~~group~~ group are large (each $n \geq 30$),
 - OR the data are relatively normally distributed in each group
 - Variability is “similar” in all ~~group~~ groups
- If these conditions are in doubt, or if the response is ordinal, then the Kruskal-Wallis test is an alternative.

$$H_0 : \text{pop median}_1 = \text{pop median}_2 = \dots = \text{pop median}_k$$

vs. $H_A : \text{At least one pair pop median}_i \neq \text{pop median}_j \text{ for } i \neq j$

- K-W test is an extension of the (Wilcoxon) rank-sum test to more than 2 groups
 - With $k = 2$ groups, the K-W test is the same as the rank-sum test

Ranks for the K-W test

1. Combine the k samples together (keep track of which observations came from each sample).
2. Rank the full set of $N = n_1 + \dots + n_k$ observations.
 - If ties exist, assign average ranks to the tied values (as with the signed-rank test).
3. Then sum the ranks within each of the k groups
 - Label the sums of the ranks for each group as $R_1, \dots, R_k$.

If H_0 is true, we expect the populations to have the same medians, and thus the ranks to be similar as well.

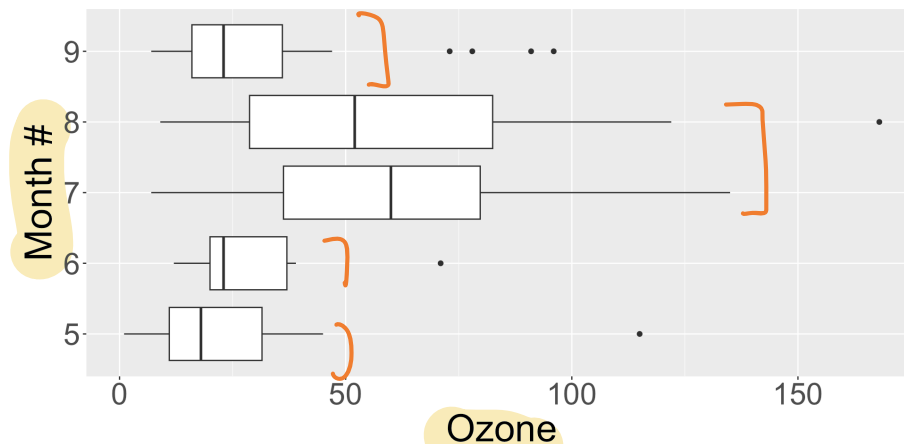
$$H_0: R_1 \approx R_2 \approx R_3 \quad k=3 \text{ (3 groups)}$$

$$H_A: R_1 \not\approx R_2 \text{ OR } R_1 \not\approx R_3 \text{ OR } R_2 \not\approx R_3$$

Example: Ozone levels by month

- `airquality` data included in base R - no need to load it
- Daily air quality measurements in New York, May to September 1973.
- Question: Do ozone levels differ by month?

Month	variable	n	mean	median	sd
5	Ozone	26	23.615	18	22.224
6	Ozone	9	29.444	23	18.208
7	Ozone	26	59.115	60	31.636
8	Ozone	26	59.962	52	39.681
9	Ozone	29	31.448	23	24.142



- Does not look like each month has equal variance so we cannot use ANOVA

```
1 max(Oz_mnth$sd) / min(Oz_mnth$sd)
```

```
[1] 2.179317
```

to use ANOVA:
 $\text{max variance of group} / \text{min var of grp} \leq 2$

K-W test in R

outcome group
↑
variable

```
1 kruskal.test(Ozone ~ Month, data = airquality)
```

Kruskal-Wallis rank sum test

data: Ozone by Month

Kruskal-Wallis chi-squared = 29.267, df = 4, p-value = 6.901e-06

$\angle 0.05 = \alpha$
reject the
null
(medians equal)

There is sufficient evidence that the median ozone levels are different in at least two months from May - September, 1973 in New York City ($p < 0.001$; Kruskal-Wallis test).

- (fyi) Since the K-W test is significant, follow-up with ~~pairwise~~ (Wilcoxon) rank-sum tests using a multiple comparison procedure to identify which months have different medians.

↓

5 v 6	6 v 7	7 v 8	8 v 9
5 v 7	6 v 8	7 v 9	
5 v 8	6 v 9		
5 v 9			